

Data Center Load & Virginia Residential Electricity Prices (2018-2025)

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BACKGROUND

- Residential electricity affordability is a live policy concern, and Virginia hosts the world’s largest data center market. Whether this new demand is reflected in household bills or absorbed elsewhere in the system is what regulators and consumers need to know.
- Data center demand has grown rapidly with the expansion of cloud computing and the Internet of Things (Koot & Wijnhoven, 2021), accelerating further after the late-2022 AI boom.
- Virginia, particularly Loudoun and Prince William counties in the Dominion (DOM) zone, has become one of the world’s largest data center markets, increasing electricity demand sharply.
- Existing work examines wholesale market effects (Kay et al., 2026; Wiser et al., 2025); the retail side, where households actually pay, remains understudied.

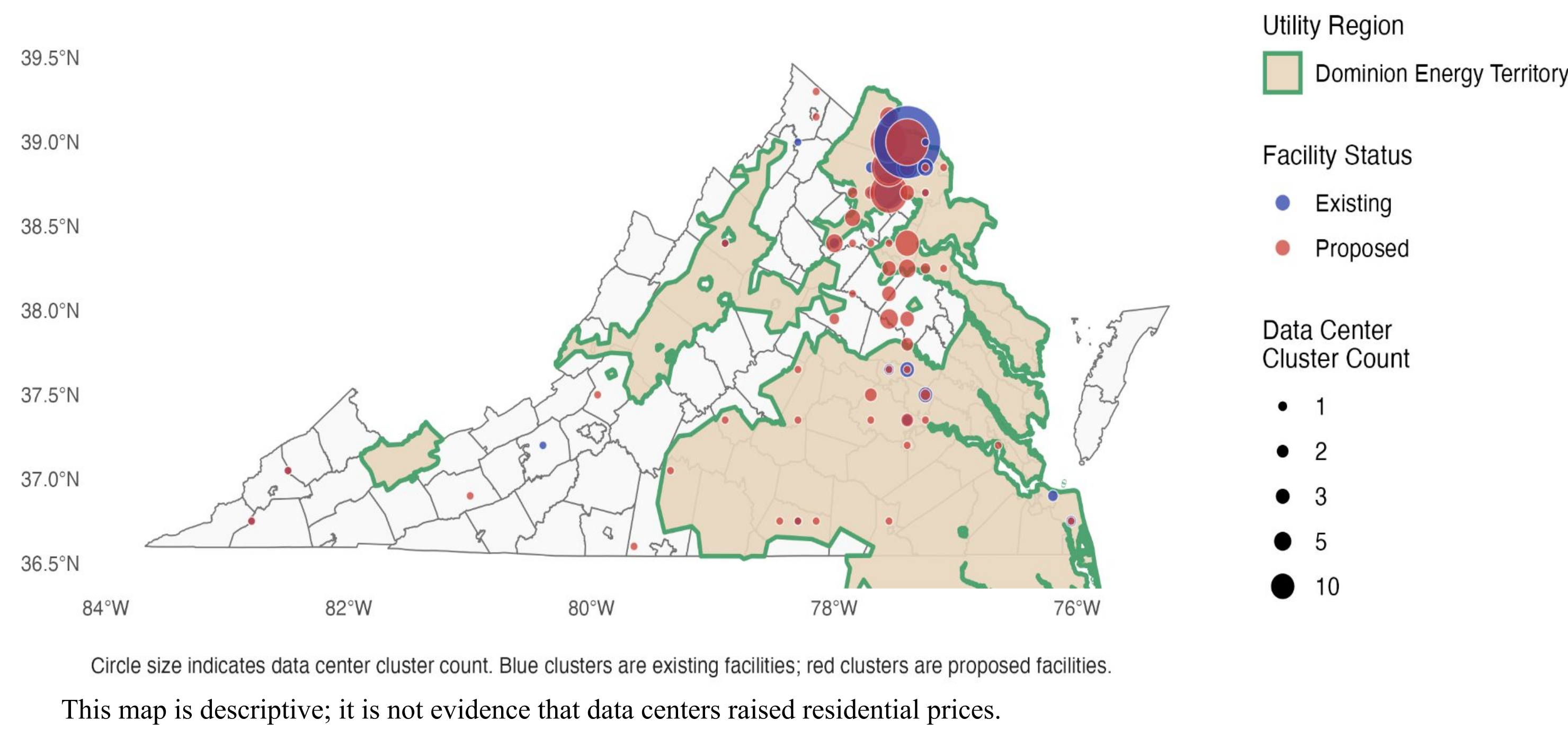
RESEARCH QUESTION

- Do the electricity demands of Virginia’s data centers show up in residential electricity bills, and has that relationship changed since the late-2022 AI boom?

DATA & METHODS

- Empirical models take data from the Energy Information Administration (EIA) and PJM Data Miner, with monthly average residential retail electricity prices being our variable of study.
- Explanatory variables include DOM load and natural gas prices as the main variables, along with Heating Degree Days (HDD), Cooling Degree Days (CDD), residential customer counts, gas generation share, and month fixed effects as controls.
- DOM zone load tracks Virginia total retail sales closely (Pearson r = 0.97), validating its use as a proxy for statewide demand.
- Figure 1 illustrates the geographic distribution of existing and proposed data centers across Virginia within the DOM, the largest PJM zone in the state, taken from ArcGIS and Cleanview Data Center Tracker.

Figure 1: Virginia Data Centers Facilities and Dominion (DOM) Zone



- Gas lags of 3, 6, and 12 months were tested; the 3- and 12-month lags are jointly insignificant ($F = 1.20$, $p = 0.31$), and the 6-month lag fits best among 6-, 8-, and 9-month alternatives (AIC 108.8 vs. 126.1, 125.6), though cross-correlation suggests the timing is approximate.
- Augmented Dickey-Fuller (ADF) and Kwiatkowski–Phillips–Schmidt–Shin (KPSS) tests confirm the series contains unit roots in levels and are stationary in differences, motivating the first-difference model and underlying why load cannot be separated from the trend.
- Both dates coincide with the natural gas price cycle rather than the data center load ramp: October 2021 falls at the onset of the 2021- 2022 gas price surge, and March 2023 at its unwind.

To estimate the relationship between data center-driven electricity demand and residential electricity prices, this study uses time-series analysis

1) Level Specification:

$$P_t = \beta_0 + \beta_1 G_{t-6} + \beta_2 L_t + \beta_3 B1_t + \beta_4 B2_t + X_t \gamma + \delta_m mon_{m(t)} + \lambda trend_t + \varepsilon_t$$

Estimates change in:

- Lagged natural gas price, G_{t-6} (real Henry Hub, \$/MMBtu, 6-month lag)
- Electricity load, L_t (DOM zone monthly load, GWh)
- Structural breaks, $B1_t$ and $B2_t$ (indicators for Oct. 2021 and Mar. 2023)
- Seasonality and trend: $mon_{m(t)}$ (month fixed effects), $trend_t$ (linear time trend)
- Load $\times$ Post interaction, $L_t \times Post_t$ (M6 only; $Post_t = 1$ from Jan. 2023)

2) First-Difference Model:

$$\Delta P_t = \beta_0 + \beta_1 \Delta G_{t-6} + \beta_2 \Delta L_t + X_t \gamma + \varepsilon_t$$

Examines whether short-term changes in electricity demand and fuel prices are associated with residential electricity prices after removing long-term trends.

- Controls, X_t (HDD, CDD, residential customer counts, gas generation share)

RESULTS

- Real Virginia residential electricity prices are associated with natural gas costs at approximately a six-month lag (0.19, $p < 0.001$), consistent with regulated fuel-factor and rate-case timing. The association with DOM zone load is positive but cannot be cleanly separated from the secular time trend, and it shows no detectable change after the late-2022 AI expansion.
- DOM Load has a statistically significant association with residential retail electricity prices in Models 2, 3, and 6 ($\beta = 0.001$, $p < 0.001$), but becomes insignificant in Models 1, 4, and 5, suggesting that the association is sensitive to the model (Table 1).

Table 1: Specification ladder: Virginia residential price

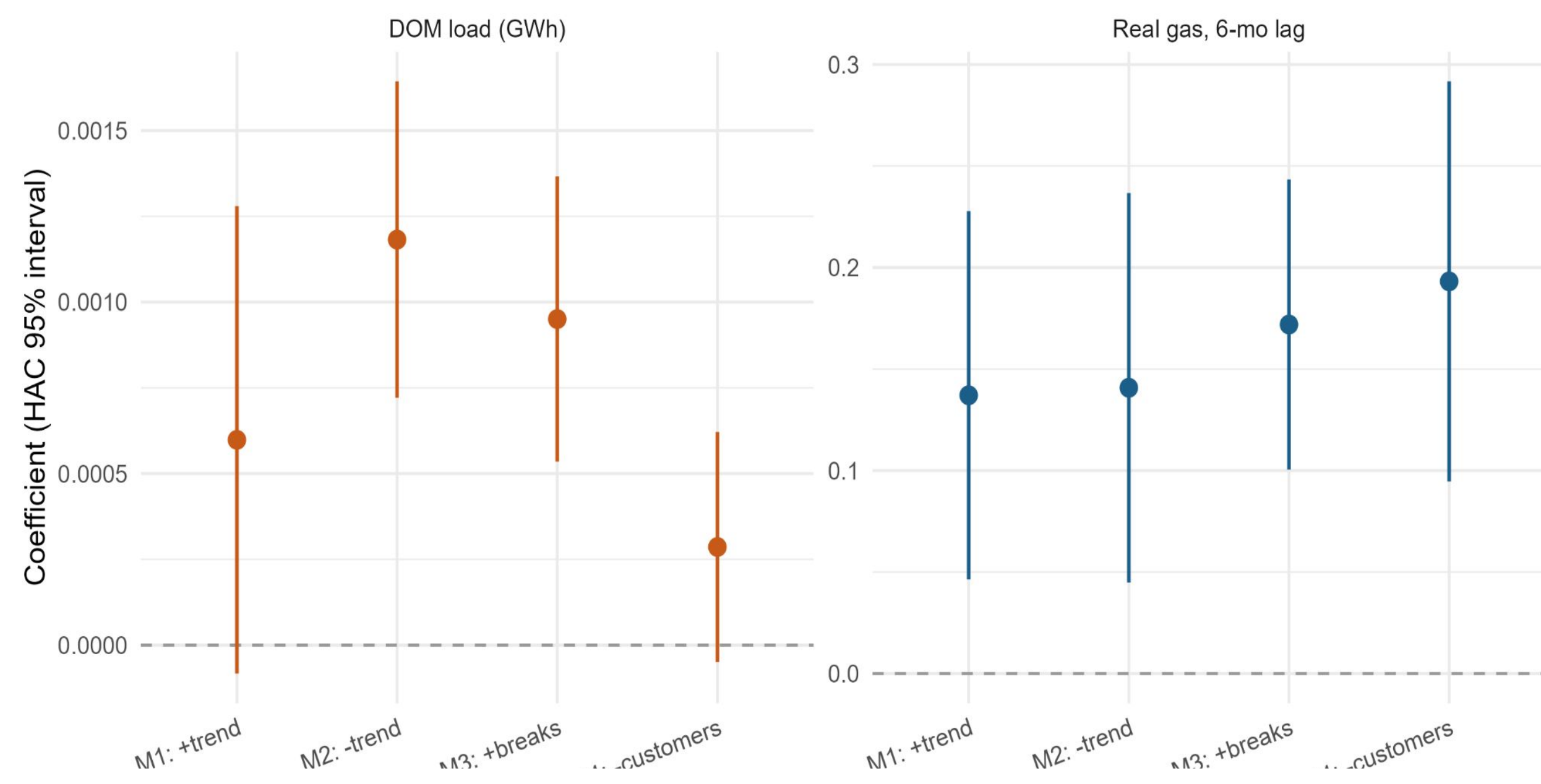
Dependent Variable: Virginia residential electricity price (cents/kWh, 2024\$)						
	M1: +trend (levels)	M2: -trend (levels)	M3: +breaks (levels)	M4: -customers (levels)	M5 (differences)	M6: interaction (levels)
DOMload (GWh)	0.001+	0.001***	0.001***	0.000+	0.000	0.001***
Post-AI (2023+)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	1.816*
Loadx post-AI						(0.703)
						-0.000+
						(0.000)
Real gas, 6-mo lag	0.137**	0.141**	0.172***	0.193***	0.142**	0.133***
	(0.046)	(0.049)	(0.036)	(0.050)	(0.053)	(0.037)
Real gas, 3-mo lag	-0.036	-0.055	0.077	0.094	-0.005	0.084
	(0.035)	(0.044)	(0.060)	(0.072)	(0.059)	(0.057)
Real gas, 12-mo lag	-0.046+	-0.037	-0.012	0.011	0.007	-0.013
	(0.026)	(0.031)	(0.026)	(0.032)	(0.037)	(0.028)
Gas generation share	6.943*	10.351***	11.935***	8.807**	4.353*	11.824***
	(3.264)	(2.889)	(2.301)	(2.948)	(1.768)	(2.231)
Residential customers count	-0.046**	-0.018***	-0.011***	0.635*	0.761*	-0.013***
	(0.014)	(0.003)	(0.003)	(0.006)	(0.006)	(0.003)
Break: Oct 2021			-1.222***	-1.625***		-1.190***
			(0.287)	(0.324)		(0.269)
Break: Mar 2023			0.635*	0.761*		
			(0.283)	(0.345)		
Linear trend	0.112*					
	(0.055)					
Num Obs	84	84	84	84	83	84
R2	0.692	0.652	0.750	0.693	0.511	0.767
R2 Adj	0.595	0.548	0.666	0.596	0.364	0.683
Std Errors	Custom	Custom	Custom	Custom	Custom	Custom

p < 0.1, *p < 0.05, **p < 0.01, ***p < 0.001.
Newey-West HAC standard errors. Month dummies included, not shown. In M5, all variables are monthly in first differences; coefficients are associations between changes. M6 tests the post-AI interaction. Break: Mar 2023 is excluded from this specification (collinear with post-AI). Coefficients are conditional associations, not causal effects.

- M3 (break-controlled, 6-month gas lag) is our preferred specification for the level relationships: it replaces an arbitrary linear trend with structural break dates located empirically, and it retains only the gas lag supported by specification testing.
- M5 (first differences) serves as a robustness check against non-stationarity. M6 tests whether the load-price association changed after the AI-driven expansion.
- The break coefficients are level shifts in the real price (-1.22 cents/kWh in October 2021 and +0.63 cents/kWh in March 2023), not changes in the load-price relationship.

- Figure 2 shows the coefficient stability across alternative models. Point estimates and 95% HAC confidence intervals compare the DOM electricity load (GWh) against a 6-month lag of real natural gas prices across models M1 to M4.
- Virginia rates are set through the State Corporation Commission (SCC) proceedings; gas costs are reflected in bills only after fuel-factor filling approval, which could explain why the load effect is hard to detect.
- Fuel shares sum to one, so only the gas share, the price relevant to marginal fuel, is included in the main model; adding solar and wind shares leaves the results unchanged.
- A single state’s series cannot isolate load from trend; a multi-state panel, contrasting data-center-heavy states with others, is the design that can, and this study’s results define exactly what it must test.

Figure 2: Coefficient Stability Across Specifications



M1-M4 differ only in co-trending controls (trend, breaks, customers). Load’s interval crosses zero when the trend or customer count is added/removed; gas stays positive and significant. Separating load from the secular trend requires cross-sectional variation, which motivates the use of a panel in the next step.

- Prices track gas costs at approximately a six-month lag; the delay is built into SCC fuel-factor and rate-case timing, and the major price shifts of 2021-2023 coincide with the gas price cycle, not data center growth.
- The load-price association cannot be separated from the general time trend, and it did not measurably change after the late-2022 AI expansion.

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